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Predicting student performance academic using Automated Machine Learning (AutoML): in medical academic institutions

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Abstract

Educational institutions seek to increase distinguished educational programs, and these programs increase costs and expenses of these institutions consuming a lot of their budget. More attention is being paid by educational institutions to keeping students in their programs without lowering educational standards or making it difficult for them to learn. Big data enables universities to predict future student outcomes more accurately. The population of the study includes students enrolled in an academic course at Damietta University between 2016 and 2021, the sample size used for the study consists of 461 student records after preprocessing, and the study was conducted at academic institution affiliated with Damietta University, Egypt. The main aim herein is to determine the best and most accurate algorithms which would enable forecasting students' academic performance, and therefore allowing medical sector college's decision makers in amending admission systems and methods in selecting students through the use of statistics and grades that have been available in the last five years.

Keywords: Machine learning, Student performance, Medical academic, Educational institutions

1 Introduction

Attracting students to join university institutions has become a pressing issue recently due to the high students'drop out rate (specifically theoretical majors).in Egypt and other nations of the world. This is why most universities and higher education institutions seek to develop a more effective strategy to solving this problem. Predicting weak students who are at risk of leaving the university, especially in the first year of the students'studies has become a necessity. The use of machine learning has become an urgent necessity to forecast not only performance but also determine students'heading towards expulsion from the university through the use of data collected during the five years preceding their studies before the beginning of their academic journey. Considering the presence of many complex models used to predict students' future data and the multiplicity of these models, arriving at the best and most accurate models is important. Using machine

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learning technology (AutoML) to arrive at the best model helps predicting the results of students to make the appropriate decisions regarding them. The current research will attempt to find the best models to predict weak students in medical colleges to help decision makers in amending admission systems and methods of selecting students for this sector.

2 Theoretical background

Over the years, Researchers academic administrators have been interested in predicting how students will do well in academic institutions. The current literature is typically studying two main issues: figuring out the most important factors for predicting student performance accurately (Rastrollo-Guerrero et al., 2020).

When it comes to figuring out what makes a student do well in school, there are many factors that that can be considered. These factors include, student's family income, student's mental and psychological state, a student's social network and the school system (Richardson et al., 2012). Feedback on the typical attributes that are utilized to foresee students' performance could be split down to internal and external factors. While, marks given for assignments, quizzes, tests, and attending classes are called internal factors, external factors include many elements such as, gender and age of students, background of a student's family or if a student has any special needs. Some scientists have also studied different factors like, students'interests, personal habits while studying and how family members support students (Khan & Ghosh, 2021).

In order to predict student'academic performance, many machine learning techniques have been employed. The following paragraphs explain how these methods are used for this purpose.

2.1 Logistic regression

Using a small set of connections between variables that are dependent and factors that are independent, logistic regression methods can predict how well students will do in school. This is done by making a function that can emulate these interactions (Strecht et al., 2015). Calculated regression is a way to predict how well students will do in school. It is often used to show the relationships between a number of independent factors that can be put into three groups: continuous, categorical, and binary (Eagle et al., 2018). The regression model can make predictions that are about 70% accurate by looking at things like career goals, cumulative grade point average (CGPA), mental scores, and personal interests.

2.2 Decision tree

Choice Tree is considered as one of the foremost well-known methods for educational information mining. The future students' academic performance was anticipated utilizing performance from past test by classifying the student's grades (Adebayo & Chaubey, 2019). Another research explores the role of behavioral attributes of students in e-learning framework. The dataset was assembled from learning administration framework. At that point, the collected features were analyzed and pre-processed in order to be ready for classification stage which is carried out by decision tree classifier (Ajibade et al., 2022). Numerous variables altogether influences the students' academic performance

such as social factors and students' behavioural activities. Thus, numerous studies propose utilizing individual, social, financial, and cognitive students' characteristics to anticipate their performance in exams (Zhao et al., 2020a).

2.3 Artificial neural network

This is a well-known classification method utilized in information mining field. In (Lau et al., 2019), conventional statistical analysis is utilized to recognize the most important factors impact the students' performance. The resulted model is assessed through numerous measurements such as, error performance, confusion matrix and error histogram. The model performance demonstrates the power of neural network features and achieves an accuracy of 84.8%. Another work looked at how input variables influenced the forecast of output classifications. It proves that the ideal input properties for the artificial neural network model were mainly focus on lectures attendance and period of study (Aydoğdu, 2020). Moreover, it may distinguish all present connection between independent variables as well as known for its superiority in detecting the correlation between independent and dependent variables (Baashar et al., 2022).

2.4 Naive Bayes

Naive Bayes classifier demonstrates its superiority in anticipating students' academic performance. It analyses all available features and makes comparisons between the independent variables in order to assess their effectiveness as predictors. Naive Bayes model is utilized to foresee students' final exams grade from diverse valuable factors such as, homework scores, lab evaluations, past exams' scores, and course participation. The model evaluation process demonstrates that the Naive Bayes performance is much way better than support vector machine, where the accomplished accuracy was 92% and 63.5%, respectively (Kaur & Bathla, 2018). Another study also concentrated on disregarded features that influence students' performance (specifically educational fee and income of the family). Utilizing Naive Bayes model demonstrates the powerful relationship between these ignored features and student' performance (Saifudin & Desyani, 2020). Majority of studies investigate utilizing Naive Bayes depend mainly on data collected from students who already started their study (Daza et al., 2022).

2.5 K- Nearest neighbours

K-Nearest Neighbour technique is known as a similarity method where is information is saved together based on their shared characteristics. In this strategy features with comparable properties are most likely gathered into the same class. To develop the predominant expectation model for students'performance, five models are examined—Naive Bayes, Decision Tree, Artificial Neural Network, K-Nearest Neighbor, and Support Vector Machine. It is proved that K-Nearest Neighbour accuracy were able to exceed all other classification models (Vital et al., 2021). To decrease processing time without affecting the accuracy percentage, a quick KNN model is proposed. Comparing the fast model against the ordinary KNN demonstrates the improvement of accuracy as it reaches 96.6%. Furthermore, the recommended fast model show decreases processing time by 90% (Ahmed et al. 2020).

2.6 Support vector machine

This is essentially what we can term as a supervised machine learning strategy that classify the dataset into its a different classes. It makes a hyperplane between two isolated classes, where elements are categorized into one class if they drop on one side of the hyperplane. The vector information point is the one closest to the hyperplane. Support vector machine was initially modelled with twofold classification in intellect, but it too consolidates a few other methods to address multi-class problems (Sen et al., 2020). For the purpose of exploring the relationship between students'social intelligent and their English exam results, Principal Component Analysis (PCA) is presented utilizing the features of support vector machine model. Exploring an interdependency among students'attributes altogether moves forward expectation results, agreeing to the expectation model (Zhao et al., 2020b). One of the foremost frequently utilized datasets in instructive inquire about is the Open University Learning Analytics (OULA) dataset which incorporates statistic data approximately understudies, demographic information, and test scores. The support vector machine model is created utilizing this dataset and the accuracy was almost 93.5% (Chui et al., 2020). When connected to diverse input features, a forecast demonstrate may give different accuracy results. As a result, the components that influence students'academic performance are pivotal within the forecast accuracy.

2.7 Ensemble of methods

There's a common agreement that integrating prediction strategies results of more robust and more exact accuracy results. The combined choice of different strategies is the result of a probabilistic averaging or a voting scheme. To expect academic performance as early as possible, an ensemble model combines variety of prediction models approaches in order to enhance the accuracy of expectation (Gil et al. 2021). This hybrid procedure merges the SVM, Naive Bayes, and Decision Tree to enhance the model accuracy. The research proves that the hybrid model acheives accuracy rate of 98.5%. In a work (Wu et al., 2019), a prediction model is proposed utilized the integration of CNN, LSTM, and SVM. It was illustrated that the hybrid model precisely predicts the result where F-measure=95.03%, whereas the SVM alone accomplishes F-measure=92.48%. Correlation-based model is utilized in (Zaffar et al. 2020) to distinguish the most important features for the prediction process, the F-measure for this model was 90%.

3 Methodology

To determine the optimum classification model and appropriate hyper-parameters for this study, we use Auto ML. Auto-Weka (Kotthoff et al., 2019) and Auto-sklearn (Feurer et al., 2015) are two of the tools with Auto ML features that are most often used. We decided to use the possibility for hyper-parameter optimization with the Auto-Weka search method. A list of predictive strategies and the associated hyper-parameter values were iterated through by an automated machine learning procedure to determine the model with the highest accuracy. This approach is depicted in Fig. 1. The search process came to the conclusion that the best classification accuracy could be achieved with an Ensemble Model consisting of several different approaches. This model was chosen

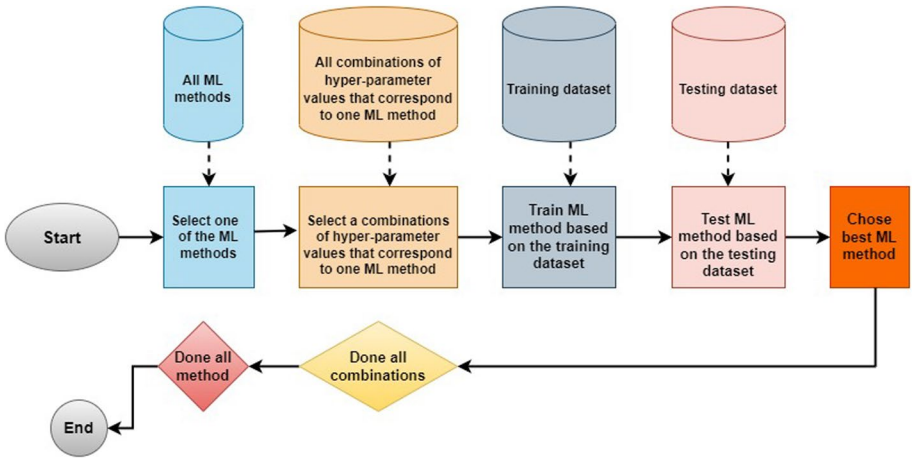


Fig. 1 Machine learning process that is automated

because it produced the best results out of all of the auto-tested pairings of prediction techniques and associated hyper-parameters. This is the forecast. The forecasting process of Ensemble Model relies on a system of voting, which chooses the anticipated results generated by the majority of its core techniques. The following are the different skills that make up the ensemble:

- Artificial Neural Network
- K-Nearest Neighbors
- Naïve Bayes
- Support Vector Machine
- Logistic Regression

3.1 Artificial neural networks (ANN)

ANN are capable of self-learning; as new information becomes available; they may improve their performance. They can be characterized as a collection of artificial neurons that represent information-based processing units, connected to one another by substantial interconnections called artificial synapses, and controlled by synaptic weight vectors and matrices. In order to model complex processes, ANN connects mathematica l"neurons"or"nodes"to create a framework. Three layers are normally present in an ANN: input, output, and hidden. The activation function of each neuron in the hidden layer and output layer might be linear or nonlinear. Sigmoid, log sigmoid, and purlin are the most often employed activation functions (Abiodun et al., 2018).

An ANN mimics a brain neuron by using the mathematical function to replicate the neural connections and interactions in the human brain. F(x) (James et al., 2013). By stacking functions based on a network model, the ANN models the connections between neurons. The n-dimensional vector [× 1, × 2,..., xn] is the parameter x for the function. We can represent the following function:

$$F(x) = S\left(\sum_{i}^{n} \omega_{\mathrm{i}} F(x_i)\right) \tag{1}$$

If x is a scalar, F(x) equals x. The factor ω, which is a weight, will be determined by the network after it has been trained on prior data. The result is normalized within a predetermined range of values using the Sigmoid transfer function. The sigmoid function was chosen because of the way that it modulates values between 0 and 1:

$$S(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

A multi-layered hierarchical model is the ANN. Each layer consists of a number of nodes (neurons) that are all joined together by unidirectional linkages to form the downstream layer's nodes. Nodes upstream or in the same layer are not connected. As depicted in Fig. 2, the standard three kinds of sections consist of a layer for input, a set of the middle levels, and a layer that provides the output. The architecture of the ANN that was used in this investigation consists of three layers: an input layer that represents all of the categorized values associated with the data characteristics that were used in this investigation, two middle layers that each have 12 and 7 neurons, and an output layer that consists of one neuron that represents the binary outcome.

3.2 K-Nearest Neighbors

Unlabeled observations are categorized using the K-Nearest Neighbors classifier by being placed in the group containing the most similar labelled samples. Each of the datasets—the training dataset, the test dataset—includes information on the observations. A data point is assigned to a category by the K-Nearest Neighbors classification method according to the predominant class of its K-nearest neighboring points in a training data set. Calculating the distance that separates two data points requires the application of a specialized function such as the Euclidean function, the Manhattan function, or the Chebychev function (James et al., 2013). The Euclidean distance function, with K set to 1, was chosen for our investigation. When two data points x and y are vectors of dimension n, then the Euclidean distance between them is as follows:

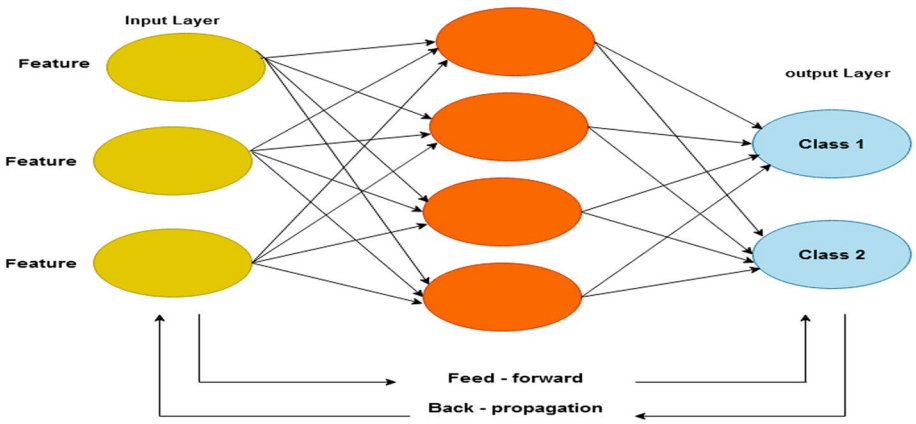


Fig. 2 ANN Hierarchy

$$E(x.y) = \sqrt{\sum_{1=0}^{n} \left(x^i - y^i\right)^2} \tag{3}$$

3.3 Naïve Bayes

One of the classification algorithms that belong to the Bayesian probabilistic family is called the Naive Bayes (NB) classifier. Because the notion that all variables tend to be conditionally independent is this classifier's most important property, we refer to it as a"Nave,"which derives from the word"nave."As a consequence of this, it is less difficult and takes less time to learn each parameter individually (Mushtaq & Mellouk, 2017). This uses the Bayes'Rule to calculate the posterior likelihood of the class as per the predictor P (c|x), as stated in Equation (Munther et al., 2020).

$$P(C\backslash X) = \frac{P(X\backslash C) * P(C)}{P(X)} \tag{4}$$

In Fig. 3, P(C) stands for the probability of the prior class, P(X|C) for the predictor's probability given the class, and P(X) for the predictor's prior probability.

3.4 Support Vector Machine

The SVM method seeks to create the best decision limit or boundary. Support Vectors, or extreme vector points, are selected in the SVM method to aid in the creation of a suitable hyperplane. The SVM algorithm is used for face detection, picture classification, text classification, and other tasks. Boundaries that lie between points of data, belonging to various classes are determined by the SVM classifier. Typically, points that fall inside specific criteria belong to a common class. Typically, the best situation is where there is a linear border that can be used to separate the data points from distinct classes. However, as demonstrated in Fig. 3, data overlaps prevent this from happening in the majority of situations. Using a particular kernel function. SVM alters these points of data by converting them to a new, augmented dimension space where they can be separated linearly using a hyperplane. If non-linearly separable data points were to be expressed

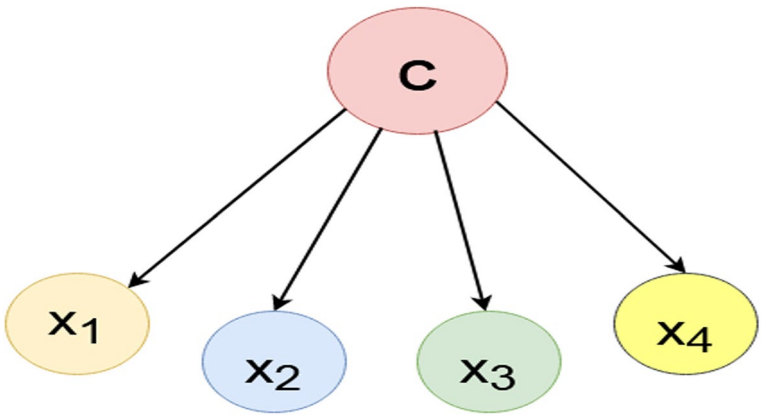


Fig. 3 Naïve Bayes network structure

onto a space with greater dimensions using a non-linear alteration, the Cover's Theorem predicts that a hyperplane is likely to be capable to divide them into two distinct groups. The bordering data points, also known as the support vectors, will be used to make the boundary hyperplane. The determined support vectors must be a certain distance from the boundary. The kernel function handles casting to a new space as well as providing the dot product between two data points, x and y, hence reducing computing complexity. In this study, we employed the polynomial kernel function F of degree d, as shown below. (Zeineddine et al., 2021). Figure 4 shows two classes of data that can be separated by a single straight line.

$$F(x.y) = \left(\sum_{i=1}^{n} x_i \bullet y_i + c\right)^d \tag{5}$$

3.5 Logistic regression

The logistic regression (LR) model is one of the most popular methods for multivariate regression analysis (Chen & Chang, 2016). It is one of the most popular methods because of its algorithmic effectiveness and its capacity to address difficult nonlinear difficulties by insertion of a suitable connecting function into the initial linear regression model. When the variable that is dependent is a variable that is categorical, and especially when the dependent variable is binary, the method may effectively carry out the classification, as well as effectively forecast the likelihood of occurrence of the dependent variable. As a consequence of this, the logistic regression classifier makes use of the logistic function L, which is demonstrated in the function shown below, to convert the outcome of a linear regression function f(x) into an outcome that falls somewhere in the range of 0 and 1 (Zhang et al., 2020).

$$\mathrm{L}_{(\mathrm{f(x)})} = \frac{1}{1 + \mathrm{e}^{-\mathrm{f(x)}}} \tag{6}$$

4 Results

4.1 Dataset description

The population of the study includes students enrolled in academic course at Damietta University between 2016 and 2021.

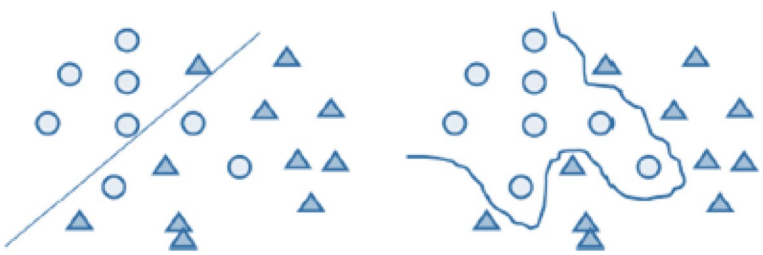


Fig. 4 Non-linearly separable overlapping data classes

Table 1 The number of students records in each academic year before and after preprocessing

Academic Year	Records before excluding outliers' values	Records after excluding outliers' values
2016	99	94
2017	22	14
2018	121	117
2019	93	92
2020	62	62
2021	83	82
Total	480	461

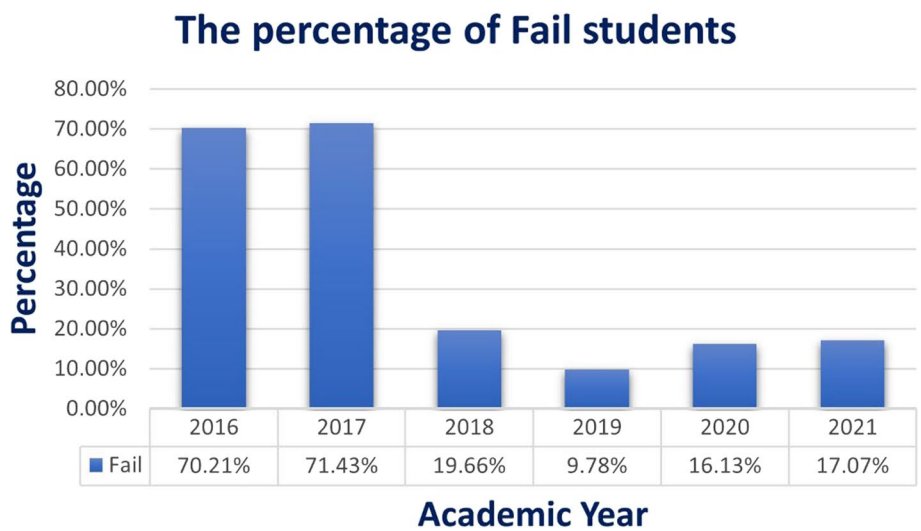


Fig. 5 The percentage of failing students in a course university during the academic years (2016–2020)

The dataset for this paper was collected from academic results of students within a faculty of Damietta University in Egypt. Particularly, we depended on students records in one university course with different features. Consequently, these records rely on different data features of students' academic results to help in generating predictive model with high accuracy. Furthermore, this predictive model in turn should expect the success grades of students in future in the university course. We encountered many obstacles in building the predictive model like inconsistency, outliers, missing, imbalanced, overlap data values.

The main dataset contains 480 records of students' academic results in one of the courses offered in the university. The dataset contains six categories based on the academic years from 2016 to 2021. The whole number of students'academic records are (480) instances without preprocessing or excluding missing, outliers and imbalance values. Table 1 shows each academic year and the related number of students before and after processing. The finish number of students'academic records in dataset is (461) instances.

The academic performance outcomes for learners in a chosen university course that are discussed in this article demonstrate the percentage of students who received a

Table 2 A comparison different grades levels of students during academic years (2016–2021)

Academic year	Total	A	%	B	%	C	%	D	%	F	%
2016	94	4	4.26	1	1.06	5	5.32	18	19.15	66	70.21
2017	14	0	0	0	0	0	0	4	28.57	10	71.43
2018	117	0	0	6	5.13	28	23.93	60	51.28	23	19.66
2019	92	2	2.17	23	25	29	31.52	29	31.52	9	9.78
2020	62	5	8.06	11	17.74	15	24.19	21	33.87	10	16.13
2021	82	1	1.22	7	8.54	17	20.73	43	52.44	14	17.07
Total	461	12	2.6	48	10.41	94	20.39	175	37.96	132	28.63

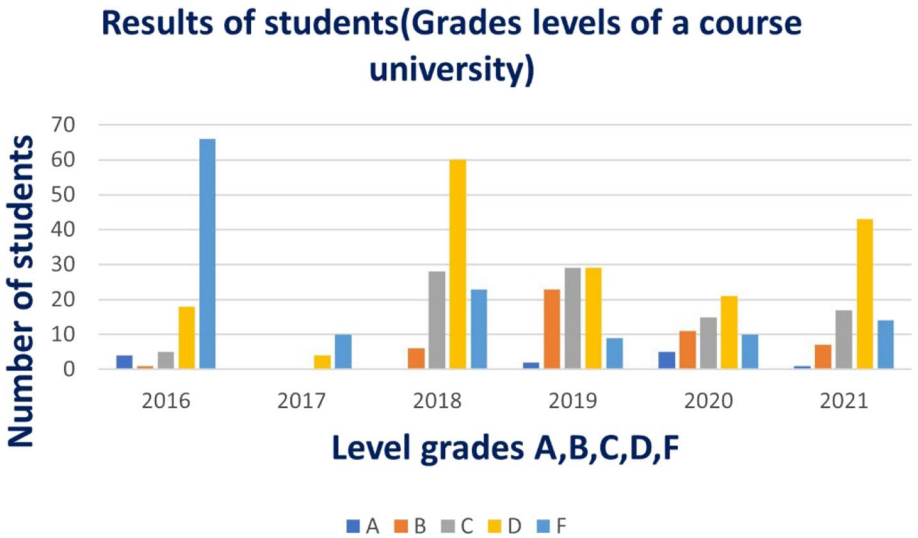


Fig. 6 Results of students (Grades levels of a course university A, B, C, D, F)

failing grade, as shown in Fig. 5. 2017 was the year in which the failure rate reached a record high of 71.21%. In the 2016 school year, the rate of failure was 70.21 percent. In the 2018 school year, the rate of students who did not pass their courses was 19.66%. In the 2019 school year, the percentage of students who did not pass their courses was 9.78%. The percentage of students who did not graduate during the school year 2020 was 16.13%. The percentage of students who did not graduate during 2021 registered at 17.07%.

In this study, the dataset was analyzed by using the frequency distribution of the categorical remarks variable, which is displayed in Table 2. The level grades of dataset from academic year 2016 to 2021 showed that (12) students have achieved grade A in percentage (2.6%). While (48) students have achieved grade B in percentage (10.41%). Besides, (94) students have achieved grade C in percentage (20.39%). Moreover, it has been observed that 175 students (37.96%) obtained a D grade while 132 students (28.63%) achieved D grade. To sum up, the percentage of D grade which expressed pass grade has achieved the high percentage followed by F grade which expressed fail grade as depicted in Fig. 6.

Figure 6 describes the graph which explained the number of students had achieved grade levels (A, B, C, D) in one of university course in academic years (2016–2021).

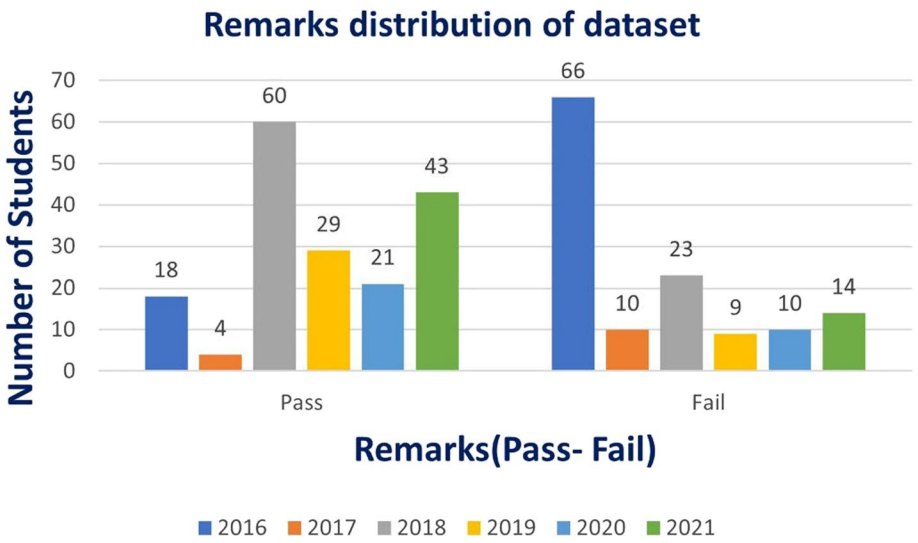


Fig. 7 A comparison between (Pass- Fail) grade levels in dataset

Table 3 Features of dataset

No	Variable	Description
1	Academic Year	Academic year from 2016 to 2021
2	Midterm	Degree of the exams through academic year from 0 to 20
3	Writing_ exam	Degree of the final exam as a writing exam which was hold at the end of studying course from 0 to 60
4	Final degree	Sum of degrees of midterm and writing exam from 0 to 100
5	Grade	Grades obtained from final total degree A: degree range [85–100] B: degree range [75–84] C: degree range [65–74] D: degree range [50–64] F: degree range [0–49]

Although the number of successful students is greater than the number of failure students, the number of failure students represents a high percentage. Fig. 7 depicted a comparison between pass grade expressed by D and fail grade expressed by F.

4.2 Dataset features

The dataset contains 461 instances after excluding 19 instances of outliers and missing values, after data collection. Dataset was divided into train (329 instances (70%)) and test (132 instances (30%)). Besides, the dataset involved 5 features as depicted in Table 1 such as Academic Year, Midterm, Writing_ exam, Final degree, and Grade. Table 3 explains the features and their descriptions.

The grade in dataset is classified into 5 categories (A, B, C, D, F) and number of students of each category through academic years [2016–2021] is depicted in Fig. 8.

4.3 Evaluation measures

One of the most significant issues in classification algorithms is how to estimate the prediction of educational data, which indicates evaluating how accurately they expect

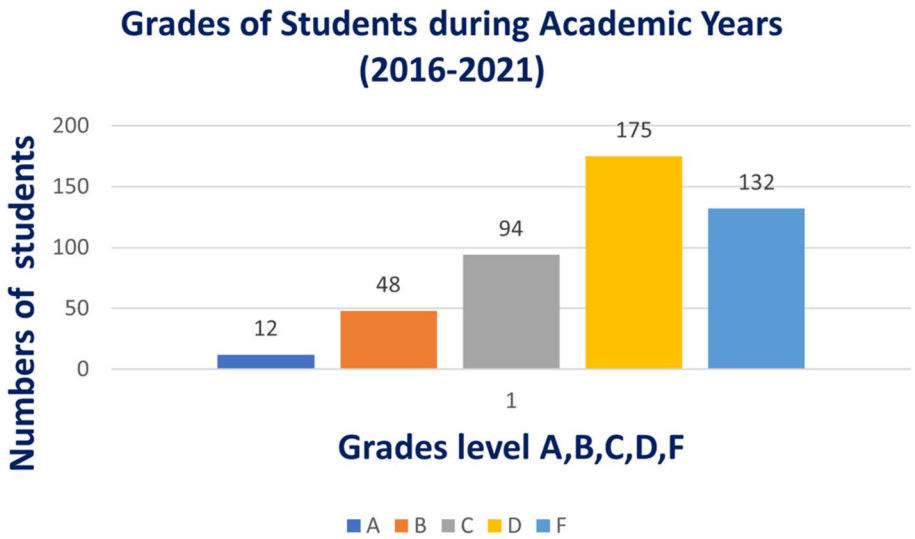


Fig. 8 Number of students in each category of grade (**A**, **B**, **C**, **D**, **F**) through academic years (2016–2021)

the desired result (Tohka & Van Gils, 2021). This section presented the metrics to estimate the classification implementation:

- *Confusion matrix (CM):* is a brief of the forecast results; the number of true and false forecasts is outlined with calculation weights and broken down by class. From this confusion matrix depicted in Table 3, TP and TN denote the number of positive and negative examples that are accurately categorized. FP and FN represent the number of misclassified negative and positive examples, respectively (Hussain et al., 2022).
- *Precision (P):* is the number of accurate classes produced by the classification measure and is estimated by this equation (Hossin & Sulaiman, 2015):

$$Precision = \frac{TP}{(TP + FP)} \tag{7}$$

- *Recall (R):* is the metric used to find all relevant instances in a dataset; it is calculated as the total quantity of positive identifications minus the number of false negatives (Hussain et al., 2022).

$$Recall = \frac{TP}{(TP + FN)} \tag{8}$$

- *F-measure:* is calculated via the following equation (Hossin & Sulaiman, 2015) and represents the harmonic mean of accuracy and recall:

$$F1 = \frac{T2 * Precision * Recall}{(Precision + Recall)} \tag{9}$$

or developed through the proceeding equation (Hossin & Sulaiman, 2015):

$$F1 = \frac{2 * TP}{(2 * TP + FP + FN)} \tag{10}$$

– *Accuracy (A):* the accuracy or is the most used evaluation measure in practice either for binary or multi-class classification problems. Accuracy determines the quality of the produced solution estimated based on the percentage of true predictions over total examples (Muntean and Militaru 2023). Accuracy is the number of true forecasts made as a ratio of all predictions made and is determined this equation (Hussain et al., 2022).

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN)} \tag{11}$$

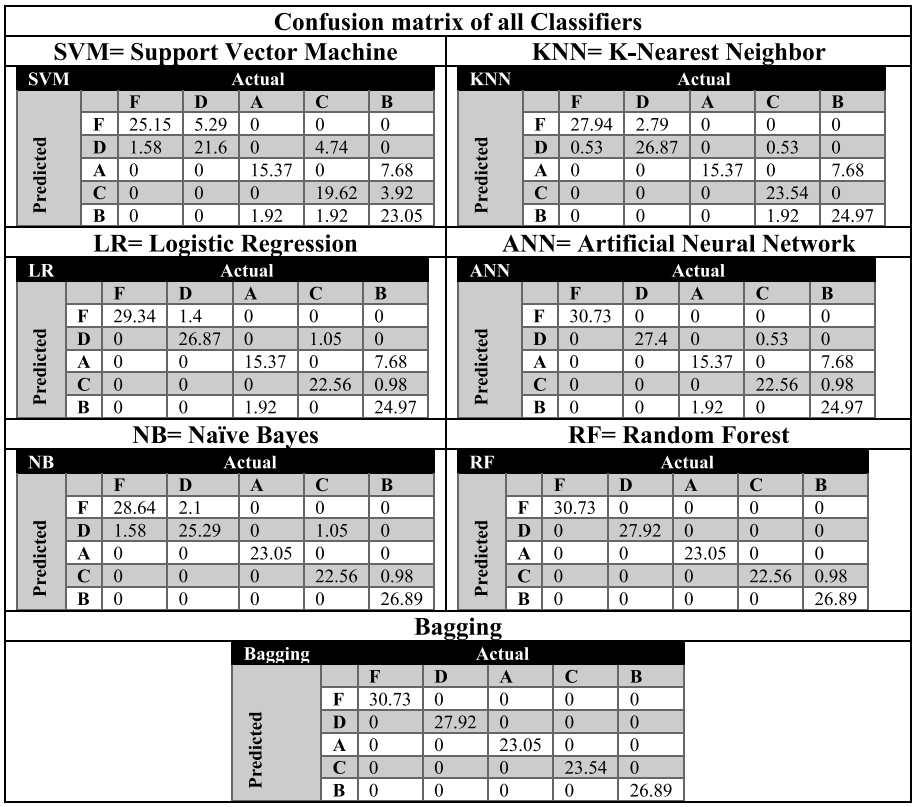
– *Classification report:* is used to estimate the rate of forecasts of a classification algorithm; it includes these measurements: precision, recall, and F1 score.
– *Kappa coefficient (κ):* which is a metric for describing how well two classifiers agree with one another. It accounts for the possibility of coincidental agreements. The following equation can be used to describe it (Muntean and Militaru 2023):

$$k = \frac{\mathrm{P0} - Pe}{1 - Pe} \tag{12}$$

Po is the probability of making the right prediction. *Pe* is the probability of accidental agreement between the classifiers. In a binary system having two predictors, $Pe = P1(a) \bullet P2(a) + P1(b) \bullet P2(b)$, where $Pi(n)$ is the probability of classifier *I* am predicting class *n*. A kappa coefficient between 0.4 and 0.75 is considered good according to Fleiss' Scale. A kappa below 0.4is poor.

4.4 Evaluation of classification methods on dataset

As mentioned before, this dataset contains (461) instances which express the academic records of students in one of courses within Damietta university. The dataset divides into training set which contains (329) instances in percentage of 70% and testing set contains (132) instances in percentage of 30%. The all instances during the academic years from 2016 to 2021. The classification methods which had been applied on predictive model are Support Vector Machine, K-Nearest Neighbors, Logistic Regression, Artificial Neural Network, Naive Bayes, Random Forest, and Bagging. In this section we compare among all classification methods using the following evaluating metrices like kappa statistics, accuracy, precision, recall, F- measure were calculated from the TP, TN, FP, and FN values of the Confusion Matrix as depicted in Fig. 9 for each classifier.



Confusion matrix of all Classifiers

SVM= Support Vector Machine

SVM		Actual				
		F	D	A	C	B
Predicted	F	25.15	5.29	0	0	0
	D	1.58	21.6	0	4.74	0
	A	0	0	15.37	0	7.68
	C	0	0	0	19.62	3.92
	B	0	0	1.92	1.92	23.05

KNN= K-Nearest Neighbor

KNN		Actual				
		F	D	A	C	B
Predicted	F	27.94	2.79	0	0	0
	D	0.53	26.87	0	0.53	0
	A	0	0	15.37	0	7.68
	C	0	0	0	23.54	0
	B	0	0	0	1.92	24.97

LR= Logistic Regression

LR		Actual				
		F	D	A	C	B
Predicted	F	29.34	1.4	0	0	0
	D	0	26.87	0	1.05	0
	A	0	0	15.37	0	7.68
	C	0	0	0	22.56	0.98
	B	0	0	1.92	0	24.97

ANN= Artificial Neural Network

ANN		Actual				
		F	D	A	C	B
Predicted	F	30.73	0	0	0	0
	D	0	27.4	0	0.53	0
	A	0	0	15.37	0	7.68
	C	0	0	0	22.56	0.98
	B	0	0	1.92	0	24.97

NB= Naïve Bayes

NB		Actual				
		F	D	A	C	B
Predicted	F	28.64	2.1	0	0	0
	D	1.58	25.29	0	1.05	0
	A	0	0	23.05	0	0
	C	0	0	0	22.56	0.98
	B	0	0	0	0	26.89

RF= Random Forest

RF		Actual				
		F	D	A	C	B
Predicted	F	30.73	0	0	0	0
	D	0	27.92	0	0	0
	A	0	0	23.05	0	0
	C	0	0	0	22.56	0.98
	B	0	0	0	0	26.89

Bagging

Bagging		Actual				
		F	D	A	C	B
Predicted	F	30.73	0	0	0	0
	D	0	27.92	0	0	0
	A	0	0	23.05	0	0
	C	0	0	0	23.54	0
	B	0	0	0	0	26.89

Fig. 9 Confusion Matrix of classification methods were applied to dataset. SVM = Support Vector Machine, KNN = K-Nearest Neighbors, LR = Logistic Regression,, ANN = Artificial Neural Network NB = Naive Bayes, and RF = Random Forest

Table 4 Confusion matrix

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	True positive (tp)	False negative (fn)
Predicted Negative Class	False positive (fp)	True negative (tn)

Table 5 Comparison among evaluation metrics of classification methods

No	Classification Methods	Correctly classified instances	Incorrectly classified instances	Precision	Recall	F-measure	Accuracy	Kappa statistics
1	SVM	105	27	0.81	0.79	0.80	79.3%	0.74
2	KNN	119	13	0.91	0.90	0.90	89.82%	0.87
3	LR	119	13	0.91	0.90	0.90	90.13%	0.88
4	ANN	121	11	0.92	0.91	0.92	91.60%	0.89
5	NB	126	6	0.96	0.96	0.96	95.68%	0.95
6	RF	131	1	0.99	0.99	0.99	99.26%	0.99
7	Bagging	132	0	1.0	1.0	1.0	100%	1.0

SVM Support Vector Machine, *KNN* K-Nearest Neighbors, *LR* Logistic Regression, *ANN* Artificial Neural Network, *NB* Naive Bayes, *RF* Random Forest

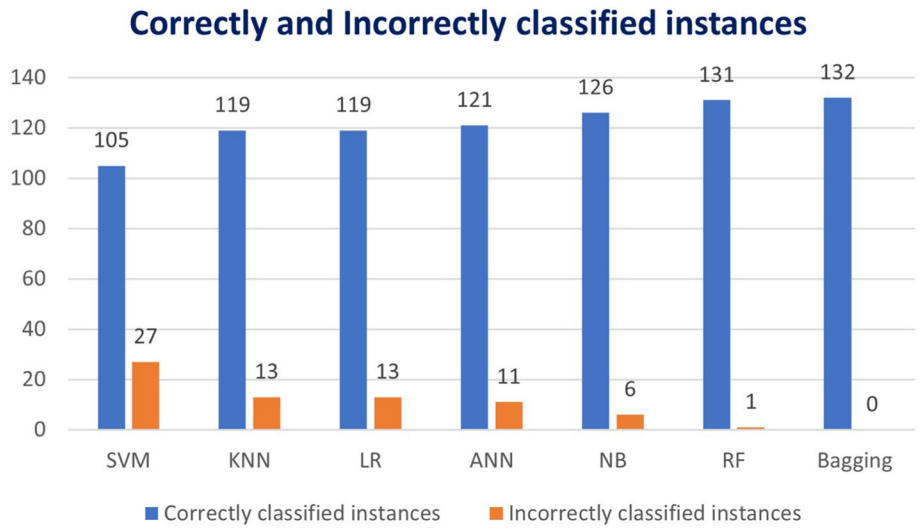


Fig. 10 Correctly vs. Incorrectly classified instances for each classifier. SVM = Support Vector Machine, KNN = K-Nearest Neighbors, LR = Logistic Regression,, ANN = Artificial Neural Network NB = Naive Bayes, and RF = Random Forest

Table 4 addresses the evaluation metrices of classification methods such as number of each correctly and incorrectly classified instances, accuracy of each classifier, precision, recall, F-measure, and kappa statistics (Table 5).

The dataset used 7 classification methods to predict the academic records of students like Support Vector Machine, K-Nearest Neighbors, Logistic Regression, Artificial Neural Network, Naive Bayes, Random Forest, and Bagging. Figure 10 depicted the numbers between correctly classified instances Vs. Incorrectly classified instances for each classifier. The ensemble methods like Random Forest, and Bagging had achieved a higher number of correctly classified instances than other classification methods. Random Forest has classified 131 correctly instances and only one incorrectly instance. While Bagging has classified all 132 instances correctly. Correspondingly, Naive Bayes has classified 126 correct instances and only 6 incorrect instances. Likewise, Artificial Neural Network has classified 121 correctly instances and only 11 incorrectly instances. While Logistic Regression and K-Nearest Neighbors have classified 119 correctly instances and only 13 incorrectly instances. Besides, Support Vector Machine has classified 105 correctly instances and only 27 incorrectly instances.

One of the important evaluations metrices is accuracy. As shown in Fig. 11 the ensemble methods have achieved higher accuracy than other classifiers. Random Forest classifier has achieved accuracy in percentage 99.26% and Bagging has achieved accuracy in percentage 100%. Likewise, Naive Bayes has achieved accuracy in percentage 95.68%. Correspondingly, Artificial Neural Network has achieved accuracy in percentage 91.6%. Similarly, Logistic Regression has achieved accuracy in percentage 90.13%. While K-Nearest Neighbors has achieved accuracy in percentage 89.82%. Besides, Support Vector Machine has achieved accuracy in percentage 79.3%.

There are other evaluation metrices of classification which can be described in Fig. 12 like precision, recall, F-measure and kappa statistics. The ensemble methods

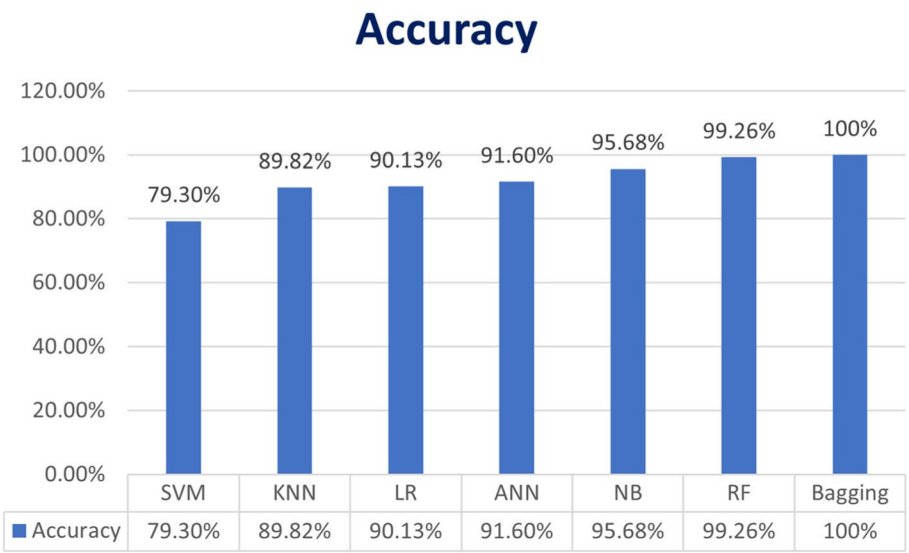


Fig. 11 Comparison accuracy among classification methods. SVM = Support Vector Machine, KNN = K-Nearest Neighbors, LR = Logistic Regression, ANN = Artificial Neural Network NB = Naive Bayes, and RF = Random Forest

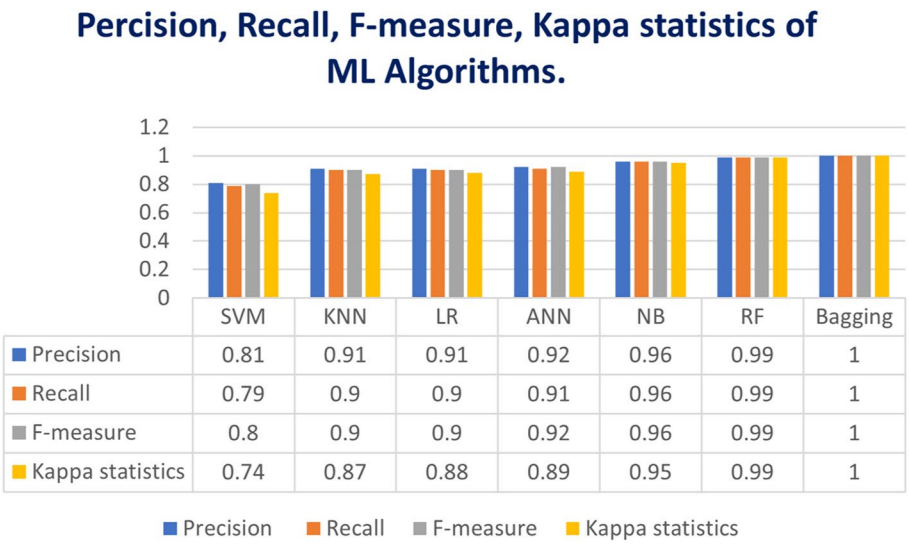


Fig. 12 Comparison of precision, recall and F-measure, kappa statistics metrices among ML classifiers. SVM = Support Vector Machine, KNN = K-Nearest Neighbors, LR = Logistic Regression, ANN = Artificial Neural Network, NB = Naive Bayes, and RF = Random Forest

have achieved high precision, recall, F-measure and kappa statistics. Random Forest has achieved precision (0.99), recall (0.99), F-measure (0.99) and kappa statistics (0.99). Similarly, Bagging has achieved precision (1), recall (1), F-measure(1) and kappa statistics(1). Likewise, Naive Bayes has achieved precision (0.96), recall (0.96), F-measure (0.96) and kappa statistics(0.95). Correspondingly, Artificial Neural Network has achieved precision (0.92), recall (0.91), F-measure (0.92) and kappa statistics (0.89). Likewise, Logistic Regression has achieved precision (0.91), recall (0.90), F-measure(0.90) and kappa statistics(0.88). Similarly, K-Nearest Neighbors has

achieved precision (0.91), recall (0.90), F-measure (0.90) and kappa statistics(0.87). Support Vector Machine has achieved precision (0.81), recall (0.79), F-measure (0.80) and kappa statistics (0.74).

5 Discussion

In this study, the RF and NP had accuracy percentages of 72.4% and 88.3%, whereas the values were 99.26% and 95.68% in the current investigation. (Ma & Zhou, 2018). Similarly, the KNN algorithm in this study was 89.82%, while it was 92.6% in Ma & Zhou (2018).

Additionally, the accuracy levels for the RF and NP in Ma and Zhou (2018) were, respectively, 72.4% and 88.3%, but in the current investigation, the values were 99.26% and 95.68%. The KNN algorithm was also 89.82% in the current study compared to 92.6% in Ingale (2021). These findings demonstrate that because of their interdependencies, when trained with all features, he ensemble methods have achieved higher accuracy than other classifiers. Random Forest classifier has achieved accuracy in percentage 99.26% and Bagging has achieved accuracy in percentage 100%. Likewise, Naive Bayes has achieved accuracy in percentage 95.68%. Correspondingly, Artificial Neural Network has achieved accuracy in percentage 91.6%. Similarly, Logistic Regression has achieved accuracy in percentage 90.13%. While K-Nearest Neighbors has achieved accuracy in percentage 89.82%. Besides, Support Vector Machine has achieved accuracy in percentage 79.3%. Additionally, in a research study, the F-measure was utilized as an evaluation indicator in order to forecast the achievement of the students (Sokkhey et al., 2020). As a result, the F-measure metric was used in the current work to effectively validate the ML algorithms trained. The cases that were accurately categorized according to the F-measure vs. instances that were incorrectly classified show how well on the validation dataset the trained ML system can predict two classes. RF, NN, and NB additionally did more effectively in F-measure validation exercises than LR and KNN did. This is because accuracy is the measure that is employed to assess the machine learning algorithms that have been trained the best. The F-measure for the RF algorithm was 90%, while the KNN algorithm had a measure of 91%. The F-measure for the RF algorithm was 99%, while it was 96% for the NP algorithm and 92% for the NN algorithm. The RF algorithm was the clear winner.

6 Conclusion and future work

As a relatively new area of research, the forecasting of academic performance in higher education institutions has become possible as a result of the proliferation of various statistical approaches. The primary objective of this study is to develop a predictive model that can unearth concealed information that can significantly contribute to an increase in the overall rate of academic achievement among students. In addition, this predictive model contributes to the enhancement of the procedures for fostering the adoption of instructional techniques and the instructional process, which in turn leads to an enrichment of the learning outcomes.

In this study, we suggest a predictive model that can be useful in medical educational institutions. The model relies on a number of classification techniques, including Support Vector Machine, K-Nearest Neighbors, Logistic Regression, Artificial Neural

Network, Naive Bayes, Random Forest, and Bagging. Ensemble methods, also known as based learners, are classification strategies that integrate two or more distinct classification approaches. The combination of techniques manage the anticipated outcomes of the used base students to achieve a higher level of performance in the model. This paper has applied Bagging and Random Forest as ensemble methods in the proposed predictive model. Ensemble methods had achieved higher accuracy than other classification methods. Bagging has achieved accuracy in percentage 100% and Random Forest has achieved accuracy in percentage 99.26%. Likewise, Naive Bayes has achieved accuracy in percentage 95.68%. Correspondingly, Artificial Neural Network has achieved accuracy in percentage 91.6%. Similarly, Logistic Regression has achieved accuracy in percentage 90.13%. While K-Nearest Neighbors has achieved accuracy in percentage 89.82%. Besides, Support Vector Machine has achieved accuracy in percentage 79.3%.

Besides, the ensemble methods have achieved high precision, recall, F-measure, and kappa statistics. Random Forest has achieved precision (0.99), recall (0.99), F-measure (0.99) and kappa statistics (0.99). Similarly, Bagging has achieved precision (1), recall (1), F-measure (1) and kappa statistics (1). Likewise, Naive Bayes has achieved precision (0.96), recall (0.96), F-measure (0.96) and kappa statistics (0.95). Correspondingly, Artificial Neural Network has achieved precision (0.92), recall (0.91), F-measure (0.92) and kappa statistics (0.89). Likewise, Logistic Regression has achieved precision (0.91), recall (0.90), F-measure (0.90) and kappa statistics (0.88). Similarly, K-Nearest Neighbors has achieved precision (0.91), recall (0.90), F-measure (0.90) and kappa statistics (0.87). Support Vector Machine has achieved precision (0.81), recall (0.79), F-measure (0.80) and kappa statistics (0.74).

These are some limitations of this paper which we intend to extend in future. This paper applied on the dataset of an academic course in medical academic institutes. The features of dataset are only 5 namely, Academic Year, Midterm, Writing_ exam, Final degree, and Grade. Moreover, this paper applied the predictive model based on some classification methods like, SVM, K-NN, LR, ANN, and NB. Also, two ensemble methods have applied namely, RF and Bagging which they achieved high level of confusion matrix, precision, accuracy, F-measure, recall, and kappa statistic.

In future, we intend to increase the number of features and instances of dataset to achieve deeply analyzing of educational data. Consequently, this can improve understanding students, determine the weak students and achieving students. In addition, deeply analyzing of educational data reduces students' failure rates. The model that predicts the performance of students in educational data mining could also be improved by applying techniques for optimization including differential evolution, genetic algorithm, and others as well. These and other strategies could be used.

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Declarations

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Not applicable.

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Competing interests
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